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Prof. Benoit Combemale (Inria)
Dr. Thomas Degueule (LaBRI)

Subject: Application for PhD Position – Agents as Run-Time Exploratory Programmers for Digital Twins

Dear Prof. Combemale and Dr. Degueule,

I am applying for the PhD position on "Agents as Run-Time Exploratory Programmers for Digital Twins" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

The concept of utilizing LLMs and autonomous agents for run-time exploratory programming perfectly matches my work in Neuro-Symbolic AI and autonomous cybersecurity. I have successfully engineered RL agents that autonomously interact with and reconfigure live network environments (the Autonomous Cyber-Shield Platform). I am highly motivated to elevate this concept, designing collaborative agents capable of live 'what-if' exploration via existing digital twin APIs.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui